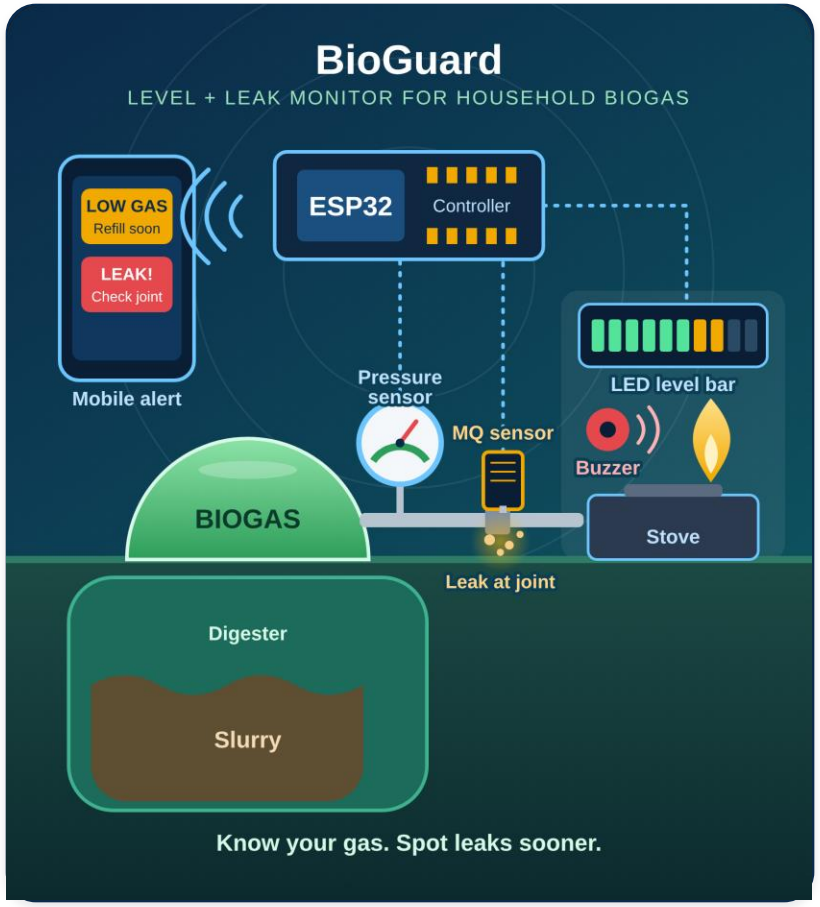
DESIGN THINKING PROJECT | 3RD YEAR

NIET Greater Noida
Autonomous Institute
Get Future Ready

BioGuard: Biogas Level & Leak Monitor

PROJECT ID	24
PROBLEM STATEMENT TITLE	Low-Cost Biogas Level and Leak Monitor for Household Digesters
PS CATEGORY	Bioenergy / Environmental Biotechnology
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ROLL NO.	2401331520036 2401331520045 2401331520060
GUIDE NAME	Ms. Barkha Bhardwaj



Concept: gas level + leak monitoring for a household digester

BIOGUARD: LEVEL & LEAK MONITOR

Proposed Solution

How might we give a household a simple, visible readout of how much biogas is available, and an early warning if gas is leaking?

Detailed explanation of the proposed solution

- Pressure sensor at the digester outlet shows gas level on an LED bar-graph
- MQ-series gas sensor near pipe joints triggers a buzzer on leak
- ESP32 / Arduino compares readings with set thresholds
- Browser dashboard simulates the full logic

How it addresses the problem

- Visible level readout: no surprise weak flame while cooking
- Early leak warning cuts wasted fuel and safety risk
- LOW and CRITICAL states warn before the gas runs out
- Optional mobile alert when level drops faster than normal use

Innovation and uniqueness of the solution

- One compact unit for both level and leak
- LED bar + buzzer works without a smartphone
- Editable thresholds and alert modes
- Built from common low-cost modules; simulator tests the UI first

TECHNICAL APPROACH

Technologies to be used

Proposed hardware

- **Low-pressure sensor:** estimates stored gas
- **MQ-series sensor:** detects combustible gas
- **Arduino / ESP32:** reads and decides
- **LED bar-graph:** level display
- **Buzzer:** leak alarm

Prototype software (built)

- **HTML5 + CSS3:** dashboard UI
- **JavaScript:** level and leak logic
- **SVG:** gauge and trend chart
- **Web Audio API:** alarm tones
- **localStorage:** saved settings

Alerts and interface

- **LED bar:** readable level from the kitchen
- **Buzzer + ALERT banner:** leak warning
- **Alert modes:** gas, leak, both or quiet
- **Optional Wi-Fi:** ESP32 mobile notification
- **Themes:** 4 selectable looks

Methodology: Design Thinking → Empathize · Define · Ideate · Prototype · Test

TECHNICAL APPROACH

Methodology and process for implementation

1 Empathize
Households cannot see stored gas

→

2 Define
Need a visible level and early leak warning

→

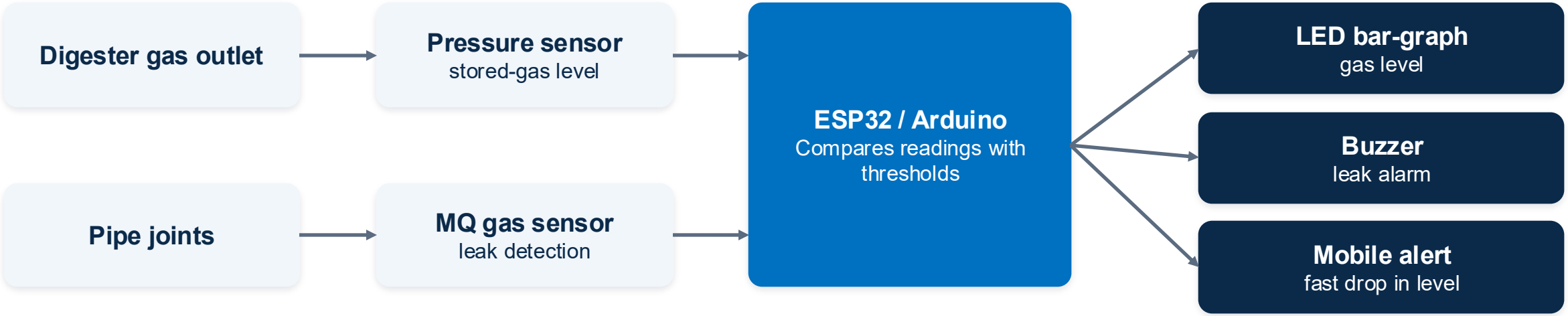
3 Ideate
Pressure gauge, MQ sensor, mobile alert

→

4 Prototype
BioGuard dashboard simulation

→

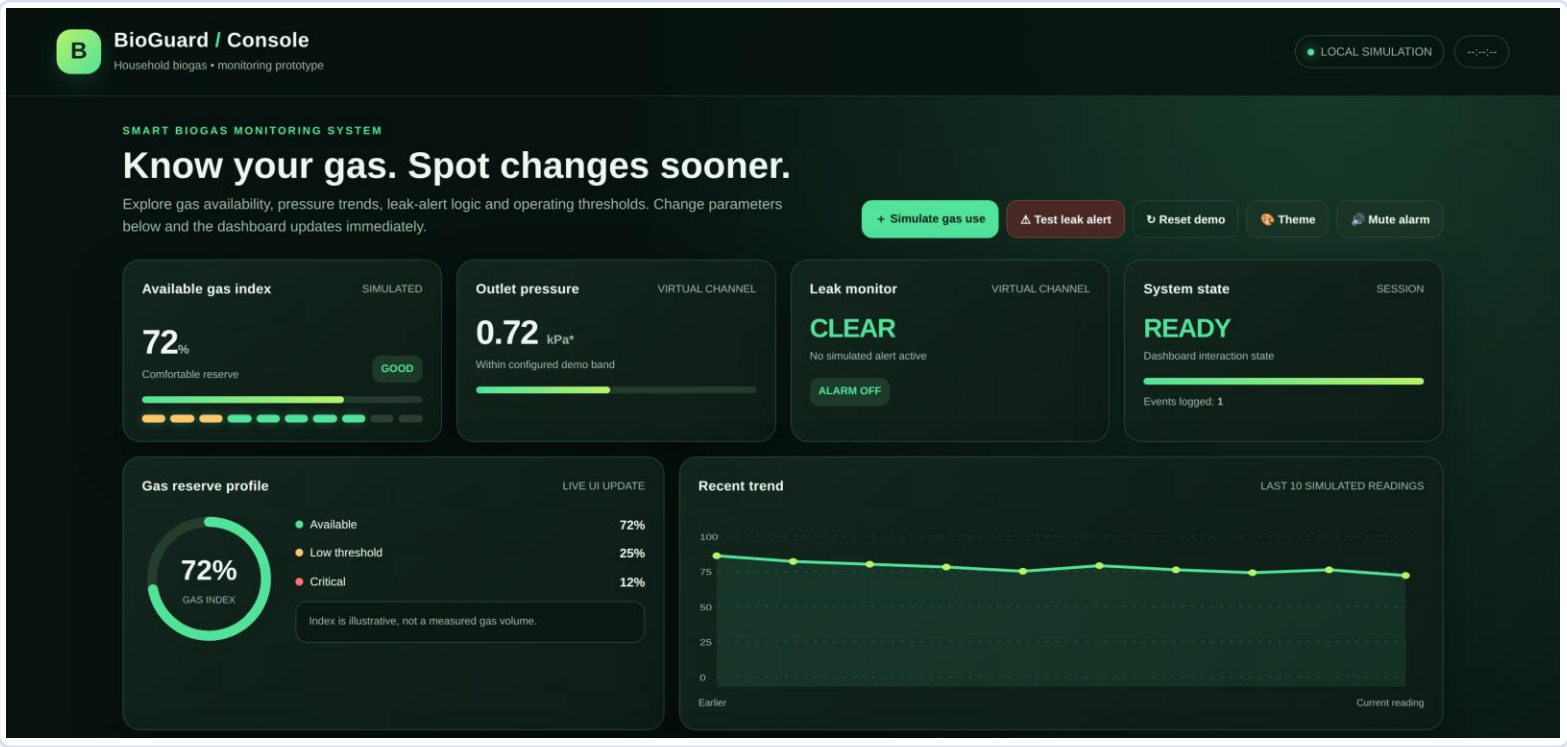
5 Test
Normal, low-gas, leak and reset scenarios



Alert logic (demo defaults): LOW ≤ 25% · CRITICAL ≤ 12% · LEAK ≥ 60 ppm* (all editable) *illustrative units

IMPLEMENTATION

Screenshots of the application



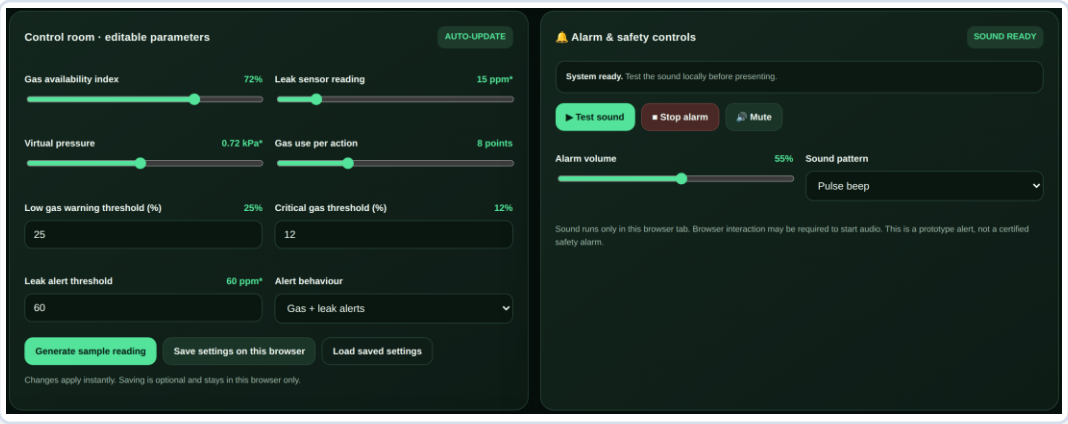
BioGuard Console: home view in normal state (gas 72% GOOD, leak CLEAR)

Quick actions: Simulate gas use · Test leak alert · Reset demo · Theme · Mute alarm

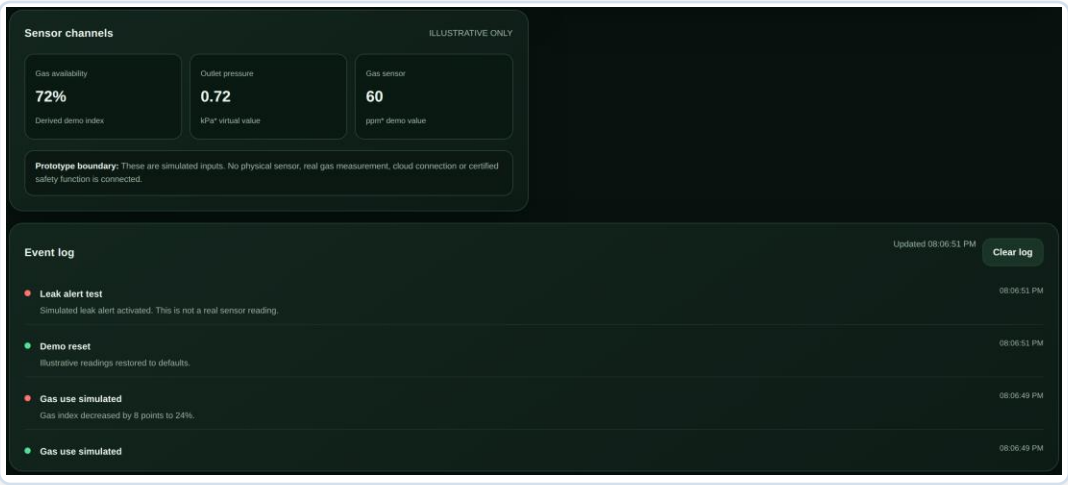
- Gas index + 10-segment LED bar
- Outlet pressure (virtual channel)
- Leak monitor: CLEAR / ALERT
- Reserve gauge with thresholds
- Trend of last 10 readings
- Event log of recent actions

IMPLEMENTATION

Key modules of the system



Control room: editable parameters + alarm and safety controls



Sensor channels (illustrative) and live event log

Sensor input layer

Virtual pressure, gas and leak channels

Gas-level engine

Maps index to GOOD / LOW / CRITICAL

Leak logic

Alert when ppm* reaches threshold

Alarm engine

Pulse, rapid or steady tone; mute

Trend and event log

10-point chart, 30-event history

Settings and themes

Editable limits, 4 themes, save

RESULTS

Performance or results obtained

Scenario	Action in dashboard	Observed output	Result
Normal use	Start state: 72% gas, 15 ppm*	GOOD · CLEAR · ALARM OFF · READY	✓ Pass
Low gas	6 × Simulate gas use (−8 each)	24%: LOW amber state, refill prompt	✓ Pass
Critical gas	8 × Simulate gas use	8%: CRITICAL and alarm tone starts	✓ Pass
Possible leak	Test leak alert, or leak at 62 ppm*	ALERT · ALARM ON · banner + tone	✓ Pass
Reset	Reset demo	Back to 72% GOOD, defaults restored	✓ Pass

5 / 5 scenarios matched the expected logic

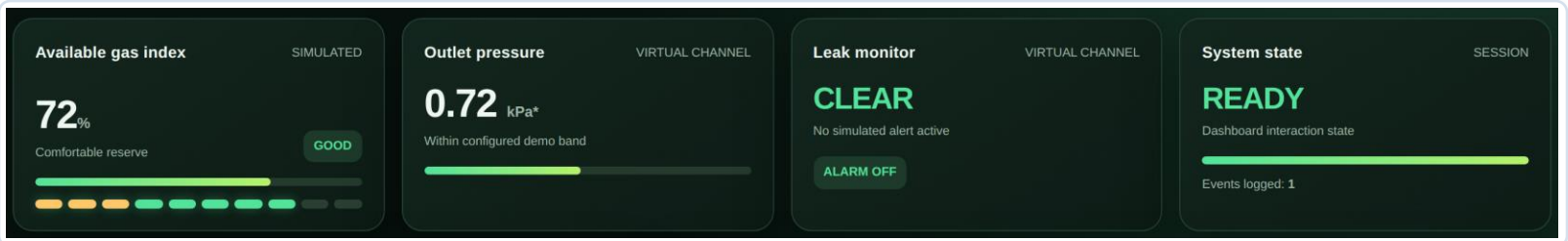
3 gas states shown: GOOD · LOW · CRITICAL

Note Readings are simulated; no physical sensor test yet

RESULTS

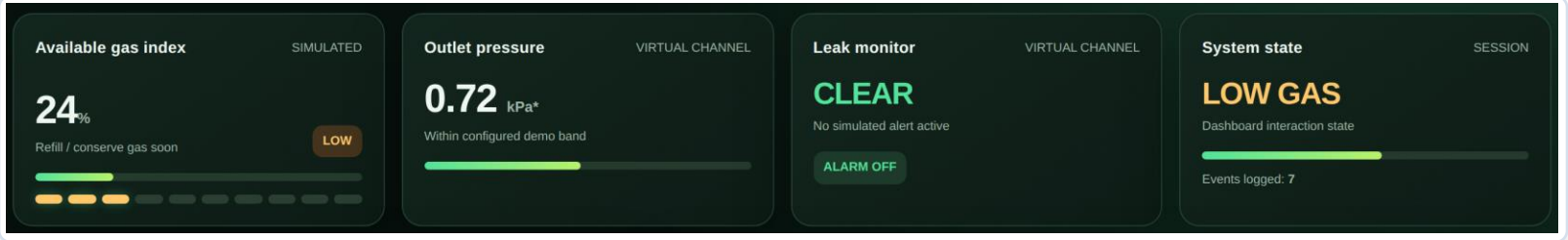
Output screens

1 Normal operation



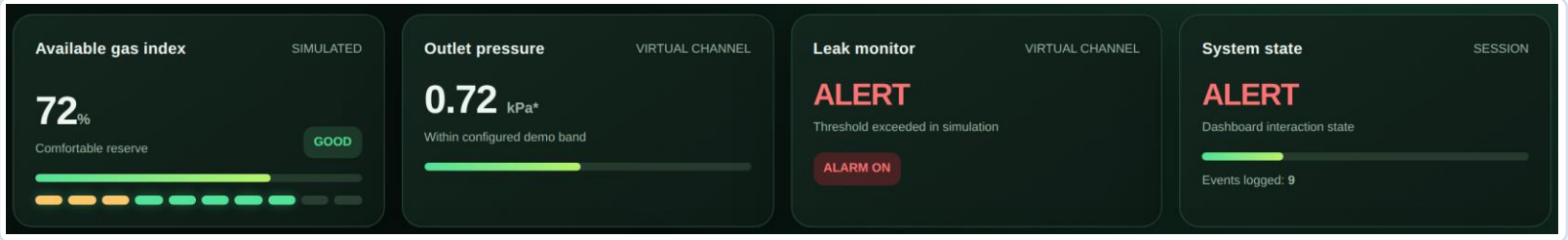
Gas 72% GOOD · Leak CLEAR · Alarm OFF · System READY

2 Low gas warning



Gas 24% LOW (amber) · refill prompt · state LOW GAS

3 Leak alert



Leak monitor ALERT · Alarm ON · red edge flash and tone

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

Technical

Common modules and controllers; bench-testable

Economic

Low-cost sensors, one controller, basic display

Operational

LED bar + buzzer; no internet needed

Potential challenges and risks	Strategies for overcoming
MQ sensor drift and cross-sensitivity	Calibrate in clean air; tune threshold with safe test gas
Pressure is not volume without calibration	Calibrate against the digester's operating range
False alarms from humidity or fumes	Warm-up time, averaging and debounce
Safe testing of flammable gas	Supervised, ventilated tests with a safe test source
Corrosion and power cuts	Protective housing and backup battery

Verdict: feasible as a bench-tested prototype with standard precautions

IMPACT AND BENEFITS

Potential impact on the target audience

4.31 M

family-type biogas plants set up in India*

*MNRE FAQ, mnre.gov.in/faq

- **Target users:** rural and semi-urban homes cooking with 1–6 m³ family plants
- **Clear answer** to “how much gas is left?”
- **Early warning** before the flame weakens or a leak grows

Social

Safer kitchens and more trust in biogas adoption

Economic

Less wasted fuel and fewer cooking interruptions

Environmental

Keeps clean cooking fuel in use; flags escaping methane

Safety

Early warning of combustible gas near pipe joints

LIMITATION

Current limitations of the project

- **Software simulation only:** no physical sensor, ESP32, LED bar or buzzer connected
- **Illustrative values:** ppm* and kPa* units are not calibrated
- **Gas index** is not a measured gas volume
- **Not a certified** gas-safety device
- **No mobile or cloud alert yet:** alerts stay in the browser tab
- **Browser limits:** audio needs a click; settings save per browser

Future scope

1. **Integrate** pressure + MQ sensors with ESP32
2. **Calibrate** pressure-to-level and leak threshold
3. **Connect** LED bar, buzzer and mobile alert
4. **Validate** safely, under supervision, in the field

RESEARCH AND REFERENCES

Details / links of the reference and research work

1. **Problem Statement 24**
   Low-Cost Biogas Level and Leak Monitor for Household Digesters (Bioenergy / Environmental Biotechnology)
   Design Thinking problem-statement sheet

2. **Hanwei Electronics: MQ-4 datasheet**
   Semiconductor sensor for natural gas / methane
   sparkfun.com/datasheets/Sensors/Biometric/MQ-4.pdf

3. **MNRE, Government of India: Biogas FAQs**
   Family-type biogas plants and national programme
   mnre.gov.in/faq

4. **Stanford d.school: Design Thinking Bootleg**
   Empathize, Define, Ideate, Prototype, Test
   dschool.stanford.edu/resources/design-thinking-bootleg

5. **Espressif Systems: ESP32 documentation**
   Microcontroller proposed for sensing and alerts
   docs.espressif.com

6. **BioGuard demo (this project)**
   Browser-based simulation of level, leak and alarm logic
   BioGuard_Demo.html